

Regression

► 4.1 Parametric Gaussian Regression

Gaussian distributions assign probability density to vectors of real numbers. In numerical applications, the objects of interest are often the output of real-valued¹ functions $f : \mathbb{X} \rightarrow \mathbb{R}$ over some input domain $\mathbb{X}$. A straightforward but powerful way to use the Gaussian inference framework on such functions is to make the assumption that f can be written as a weighted sum over a finite number F of *feature* functions $[\phi_i : \mathbb{X} \rightarrow \mathbb{R}]_{i=1,\dots,F}$, as

$$f(x) = \sum_{i=1}^F \phi_i(x) w_i =: \Phi_x^\top \mathbf{w} \quad \text{with} \quad \mathbf{w} \in \mathbb{R}^F. \quad (4.1)$$

The right-hand side of this equation introduces a slightly sloppy but helpful notation: Given a collection² $X \subset \mathbb{X}$ with elements $[X]_i = x_i \in \mathbb{X}, i = 1, \dots, N$, we will denote by $\Phi_X \in \mathbb{R}^{F \times N}$ the matrix of feature vectors with elements $[\Phi_X]_{ij} = \phi_i(x_j)$. Similarly, we will write $\mathbf{f}_X$ for the vector of function values $[f(x_1), \dots, f(x_N)]$.

The crucial algebraic property of Eq. (4.1) is that it models f as a *linear* function of the weights $\mathbf{w}$. It is therefore called a *linear regression* model.³ To perform inference on such a function within the Gaussian framework, we assign a Gaussian density over the possible values for the weight vector $\mathbf{w}$, written

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w}; \boldsymbol{\mu}, \Sigma).$$

Let us assume that it is possible to collect observations $\mathbf{y} := [y_1, \dots, y_N] \in \mathbb{R}^N$ of f corrupted by Gaussian noise with covariance $\Lambda \in \mathbb{R}^{N \times N}$ at the locations X (Figure 4.1). That is, according to the likelihood function⁴

¹ The formalism can be extended to complex-valued and multivariate functions, or to functions mapping virtually any input domain $\mathbb{X}$ to an output domain $\mathbb{Y}$ that can be made isomorphic to a real vector space. This generality is suppressed here to allow a more accessible presentation. The type of $\mathbb{X}$ does not matter at all since it is “encapsulated” and mapped to a real vector by the features ϕ . To capture multivariate *outputs* (or outputs that are isomorphic to the multivariate reals), the model has to return not one but several real numbers. This can be achieved by increasing the number of features and weights, i.e. by “stacking” several univariate functions. Chapter III, on linear algebra, prominently features this kind of model.

² We use the capital X instead of x to signify that X can really be an ordered collection of just about any type of inputs: real numbers, but also strings, graphs, etc. See e.g. §4.4 in Rasmussen and Williams (2006).

³ This need *not* mean that f is also a linear function in x !

⁴ This includes the special case $\Lambda \rightarrow 0$, usually written suggestively with the Dirac delta as

$$p(\mathbf{y} | f) = \delta(\mathbf{y} - \mathbf{f}_X).$$

This corner case is important in numerical applications where function values can be computed “exactly”, i.e. up to machine precision.

```

1 procedure PARAMETRIC_INFER( $\mathbf{y}, X, \Lambda, \phi, \mu, \Sigma$ )
2    $\Phi_X = \phi(X)$  // evaluate features of data
3    $\tilde{\Sigma} = (\Sigma^{-1} + \Phi_X \Lambda^{-1} \Phi_X^\top)^{-1}$  // posterior covariance
4   // This is the most costly step, at  $\mathcal{O}(F^3)$ .
5    $\tilde{\mu} = \tilde{\Sigma}(\Sigma^{-1}\mu + \Phi_X \Lambda^{-1}\mathbf{y})$  // posterior mean
6 end procedure

```

$$p(\mathbf{y} | f) = \mathcal{N}(\mathbf{y}; f_X, \Lambda). \quad (4.2)$$

Then the posterior over the weights $\mathbf{w}$, by Eq. (3.13), is

$$\begin{aligned}
 p(\mathbf{w} | \mathbf{y}) &= \mathcal{N}(\mathbf{w}; \tilde{\mu}, \tilde{\Sigma}), \\
 \text{with } \tilde{\Sigma} &:= (\Sigma^{-1} + \Phi_X \Lambda^{-1} \Phi_X^\top)^{-1} \\
 \text{and } \tilde{\mu} &:= \tilde{\Sigma}(\Sigma^{-1}\mu + \Phi_X \Lambda^{-1}\mathbf{y}).
 \end{aligned}$$

Using the matrix inversion lemma (Eq. (15.9)), these expressions can also be written with the inverse of a matrix in $\mathbb{R}^{N \times N}$ as

$$\begin{aligned}
 \tilde{\Sigma} &= \Sigma - \Sigma \Phi_X (\Phi_X^\top \Sigma \Phi_X + \Lambda)^{-1} \Phi_X^\top \Sigma, \\
 \tilde{\mu} &= \mu + \Sigma \Phi_X (\Phi_X^\top \Sigma \Phi_X + \Lambda)^{-1} (\mathbf{y} - \Phi_X^\top \mu).
 \end{aligned}$$

Rasmussen and Williams (2006) call this the *weight-space view* of regression. Alternatively, one can also construct a *function-space view*, a density directly over values of the unknown f : Since f is a linear map of $\mathbf{w}$, Eq. (3.4) implies that the posterior over function values f_x at a finite subset $x \subset \mathbb{X}$ is

$$\begin{aligned}
 p(f_x) &= \mathcal{N}(f_x; \Phi_x^\top \tilde{\mu}, \Phi_x^\top \tilde{\Sigma} \Phi_x) \quad \text{with} \quad (4.3) \\
 \Phi_x^\top \tilde{\mu} &= \Phi_x^\top \mu + \Phi_x^\top \Sigma \Phi_X (\Phi_X^\top \Sigma \Phi_X + \Lambda)^{-1} (\mathbf{y} - \Phi_X^\top \mu), \\
 \Phi_x^\top \tilde{\Sigma} \Phi_x &= \Phi_x^\top \Sigma \Phi_x - \Phi_x^\top \Sigma \Phi_X (\Phi_X^\top \Sigma \Phi_X + \Lambda)^{-1} \Phi_X^\top \Sigma \Phi_x.
 \end{aligned}$$

```

1 procedure PARAMETRIC_PREDICT( $x, \tilde{\mu}, \tilde{\Sigma}$ ) // predict at  $x$ 
2    $\Phi_x = \phi(x)$  // features of  $x$ 
3    $m_x = \Phi_x^\top \tilde{\mu}$  // predictive mean
4    $V_x = \Phi_x^\top \tilde{\Sigma} \Phi_x$  // predictive covariance
5    $s_x = \Phi_x^\top \cdot \text{CHOL}(\tilde{\Sigma})^\top \cdot \text{RANDNORMAL}(F) + m_x$  // sample
6 end procedure

```

Algorithm 4.1: Basic implementation of the *inference* step of parametric regression, which constructs a posterior distribution on the weights. Here, inference is performed in weight space, so the computational cost, given N observations and F features, is $\mathcal{O}(NF^2 + F^3)$. Compare with Algorithm 4.3 for the function space view, where the cost is $\mathcal{O}(N^3)$.

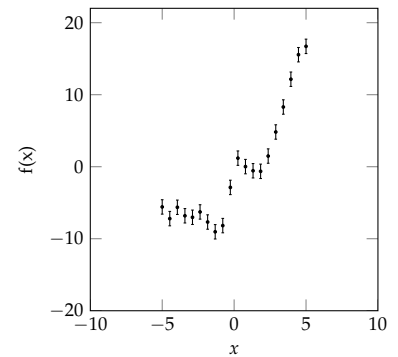


Figure 4.1: A simple nonlinear data set with i.i.d. Gaussian observation noise, i.e. $\Lambda = \sigma^2 I_N$, $\sigma \in \mathbb{R}_+$ (error-bars). Figures 4.2 and 4.3 demonstrate the effect of varying prior assumptions on the posterior induced by these observations.

Algorithm 4.2: Basic implementation of the *prediction* step of parametric regression in function space, using the outputs of Alg. 4.1. The cost, given D prediction locations, is $\mathcal{O}(DF)$ for the mean and element-wise variance, and $\mathcal{O}((DF)^2)$ for a full cov. Samples cost $\mathcal{O}((DF)^2)$ if the Cholesky decomposition (p. 36) of $\tilde{\Sigma}$ is available (e.g. from Alg. 4.1).

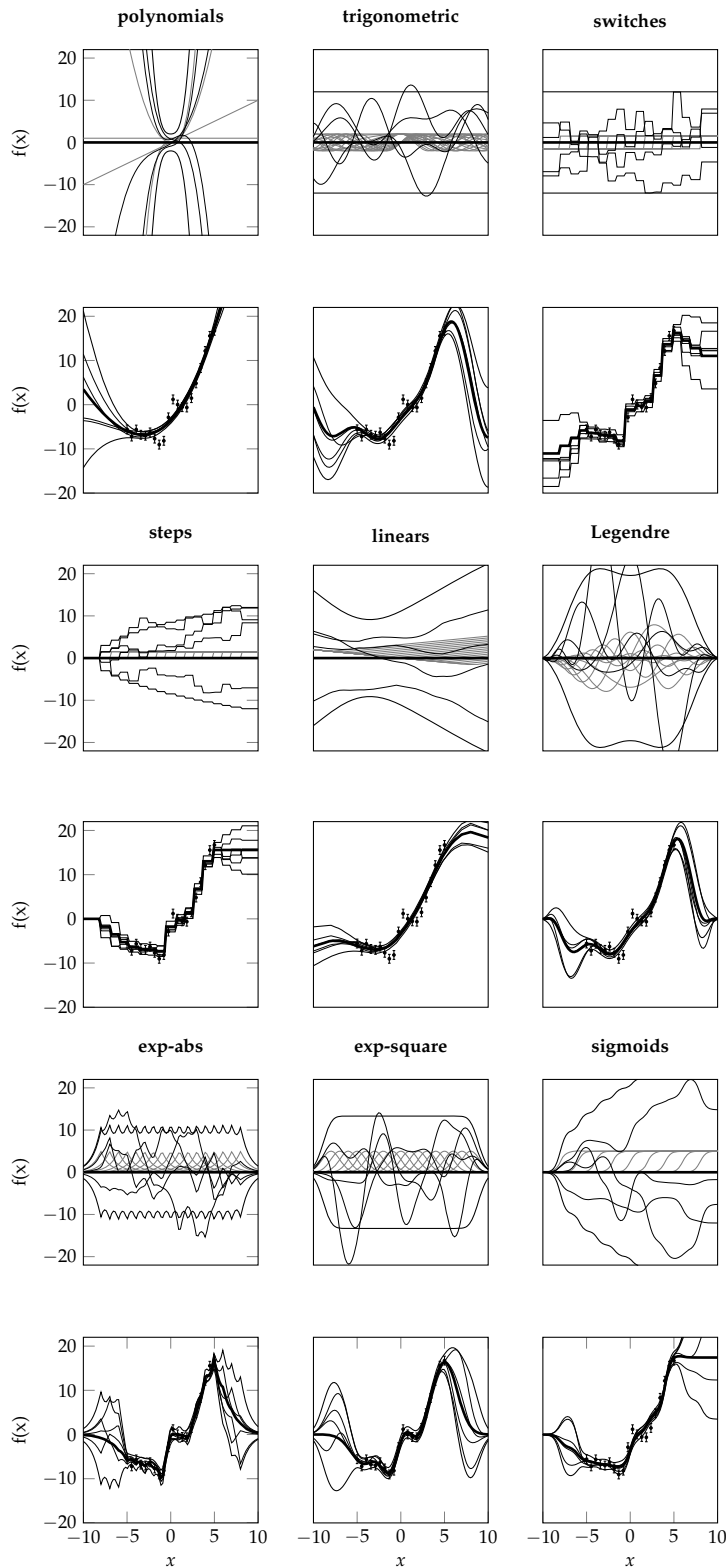


Figure 4.2: Odd-numbered rows: Priors over univariate real functions using different features. Each plot shows the underlying features Φ_x as thin grey lines, the prior mean function $\Phi_x^T \mu = 0$ arising from $\mu = 0$, the marginal standard deviation $\text{diag}(\Phi_x^T \Sigma \Phi_x)^{1/2}$ for the choice $\Sigma = I$, and four samples drawn i.i.d. from the joint Gaussian over the function.

Even-numbered rows: Posteriors arising in these models from the observations shown in Figure 4.1. The feature functions giving rise to these eight different plots are the polynomials

$$\phi_i(x) = x^i, \quad i = 0, \dots, 3;$$

the trigonometric functions

$$\phi_i(x) = \sin(x/i), \quad i = 1, \dots, 8, \text{ and}$$

$$\phi_i(x) = \cos(x/i-8), \quad i = 9, \dots, 16;$$

as well as, for $i = -8, -7, \dots, 8$, the “switch” functions

$$\phi_i(x) = \text{sign}(x - i);$$

the “step” functions

$$\phi_i(x) = \mathbb{I}(x - i > 0);$$

the linear functions

$$\phi_i(x) = |x - i|;$$

the first 13 Legendre polynomials (scaled to $[-10, 10]$); the absolute’s exponential

$$\phi_i(x) = e^{-|x-i|};$$

the square exponential

$$\phi_i(x) = e^{-(x-i)^2};$$

and the sigmoids

$$\phi_i(x) = 1/(1 + e^{-3(x-i)}).$$

The plots highlight the broad range of behaviour accessible by varying the choice of feature functions.

Figure 4.2 shows a gallery of prior and posterior densities over function values⁵ arising from the same weight-space prior $p(\mathbf{w}) = \mathcal{N}(\mathbf{w}; \mathbf{1}, I)$, and differing choices of feature functions ϕ . The figure also shows posterior densities arising from these priors under one particular data set of observations (shown in Figure 4.1 for reference). The figure and derivations above highlight the following aspects:

- ⊖ Generalised linear regression allows inference on real-valued functions over arbitrary input domains.
- ⊖ By varying the feature set Φ , broad classes of hypotheses can be created. In particular, Gaussian regression models can model nonlinear, discontinuous, even unbounded functions. There are literally *no limitations* on the choice of $\phi : \mathbb{X} \rightarrow \mathbb{R}$.
- ⊖ Neither the posterior mean nor the covariance over function values (Eq. (4.3)) contain “lonely” feature vectors, but only inner products of the form $k_{ab} := \Phi_a^T \Sigma \Phi_b$ and $m_a := \Phi_a^T \mu$. For reasons to become clear in the following section, these quantities are known as the *covariance function* $k : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ and *mean function* $m : \mathbb{X} \rightarrow \mathbb{R}$, respectively.

⁵ With some technical complications, Eq. (4.3) assigns a measure over an infinite-dimensional space of functions. But the plots only involve finitely many function values (albeit on a grid of high resolution), for which it suffices to consider the less tricky concept of a probability density. For more, see §4.2.2.

Exercise 4.1 (easy). Consider the likelihood of Eq. (4.2) with the parametric form for f of Eq. (4.1). Show that the **maximum-likelihood estimator** for w is given by the ordinary least-squares estimate

$$w_{ML} = (\Phi_X \Phi_X^T)^{-1} \Phi_X y.$$

To do so, use the explicit form of the Gaussian pdf to write out $\log p(y | X, w)$, take the gradient with respect to the elements $[w]_i$ of the vector w and set it to zero. If you find it difficult to do this in vector notation, it may be helpful to write out $\Phi_X^T w = \sum_i w_i [\Phi_X]_{:,i}$, where $[\Phi_X]_{:,i}$ is the i th column of Φ_X . Calculate the derivative of $\log p(y | X, w)$ with respect to w_i , which is scalar.

► 4.2 Gaussian Processes – Nonparametric Gaussian Inference

The basic properties of Gaussian distributions summarised in Eqs. (3.12) and (3.13) mean that computing the marginal of a high-dimensional Gaussian distribution involves only quantities of the dimensionality of the marginal. Thus one may wonder what happens if the high-dimensional distribution is in fact of *infinite* dimensionality. This limit is known as a *Gaussian process*, and extends the notion of regression to real-valued functions.

The theory and practical use of Gaussian processes are well-studied. Extended introductions can be found in the textbook by Rasmussen and Williams (2006) as well as the older book by Wahba (1990). The regularity⁶ and extremal behaviour⁷ of samples from a Gaussian process have been analysed in detail. As will briefly be pointed out below, Gaussian process models are also closely related to kernel methods in machine learning, in particular to kernel ridge regression. Many of the theoretical concepts in that area transfer – directly or with caveats – to the Gaussian process framework. These are discussed for example in the books by Schölkopf and Smola (2002), and Steinwart and Christmann (2008). Moreover, Berg, Christensen, and Ressel

⁶ Adler (1981)

⁷ Adler (1990)

(1984) provide an introduction to the theory of positive definite functions, which are a key concept in this area (see §4.2.1). With a focus on accessibility, this section provides the basic notions and algorithms required for inference in these models, as well as an intuition for the generality and limitations of Gaussian process regression.

▷ 4.2.1 Positive Definite Kernels

The main challenge in defining an “infinite-dimensional Gaussian distribution” is how to describe the infinite limit of a covariance matrix. Covariance matrices are symmetric positive definite;⁸ the extension of this notion to operators is called a positive definite kernel. We approach this area from the parametric case studied in §4.1: recall from Eq. (4.3) that, for general linear regression using features ϕ , the mean vector and covariance matrix of the posterior on function values does not contain isolated, explicit forms of the features. Instead, for finite subsets $a, b \subset \mathbb{X}$ it contains only projections and inner products, of the form

$$m_a := \Phi_a \mu \quad \text{and} \quad k_{ab} := \Phi_a^T \Sigma \Phi_b = \sum_{ij=1}^F \phi_i(a) \phi_j(b) \Sigma_{ij}.$$

These two types of expressions are themselves functions, called

$$\begin{array}{ll} \text{the mean function} & m : \mathbb{X} \rightarrow \mathbb{R}, \text{ and} \\ \text{the covariance function} & k : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}. \end{array}$$

Because the features are thus “encapsulated”, we may wonder whether one can construct a Gaussian regression model without explicitly stating a specific set of features. For the mean function, this is easy: since one can add an arbitrary linear shift b in the likelihood (4.2), the mean function does not actually have to be an inner product of feature functions. It can be chosen more generally, to be *any* computationally tractable function $m_a : \mathbb{X} \rightarrow \mathbb{R}$. The covariance function requires more care; but if we find an analytic short-cut for the sum of F^2 terms for a smart choice of (ϕ, μ, Σ) , it is possible to use a very large, even infinite set of features. This is known as “the kernel trick”.⁹ It is based on one of the basic insights of calculus: Certain structured sums and series have analytic expressions that allow computing their value without actually going through their terms individually.

⁸ For more on symmetric positive definite matrices, see §15.2.

⁹ Schölkopf (2000)

MacKay (1998) provides an intuitive example. Consider $\mathbb{X} = \mathbb{R}$, and decide to regularly distribute Gaussian feature functions

with scale $\lambda \in \mathbb{R}_+$ over the domain $[c_{\min}, c_{\max}] \subset \mathbb{R}$. That is, set

$$\phi_i(x) = \exp\left(-\frac{(x - c_i)^2}{\lambda^2}\right), \quad c_i = c_{\min} + \frac{i-1}{F}(c_{\max} - c_{\min}).$$

We also choose, for an arbitrary scale $\theta^2 \in \mathbb{R}_+$, the covariance

$$\Sigma = \frac{\sqrt{2}\theta^2(c_{\max} - c_{\min})}{\sqrt{\pi}\lambda F} I,$$

and set $\mu = 0$. It is then possible to convince oneself¹⁰ that the limit of $F \rightarrow \infty$ and $c_{\min} \rightarrow -\infty$, $c_{\max} \rightarrow \infty$ yields

$$k(a, b) = \theta^2 \exp\left(-\frac{(a - b)^2}{2\lambda^2}\right). \quad (4.4)$$

This is called a *nonparametric* formulation of regression, since the parameters w of the model (regardless of whether their number is finite or infinite) are not explicitly represented in the computation. The function k constructed in Eq. (4.4), if used in a Gaussian regression framework, assigns a covariance of full rank to arbitrarily large data sets. Such functions are known as (positive definite) *kernels*.

Definition 4.2 (Kernel). A positive (semi-) definite kernel is a bivariate real function $k : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ over some space $\mathbb{X}$ such that, for any set $X = [x_1, \dots, x_N] \subset \mathbb{X}$, the matrix k_{XX} with elements

$$[k_{XX}]_{ij} = k(x_i, x_j)$$

is positive (semi-) definite.¹¹ Such functions are also known as Mercer kernels, positive definite functions, and, in the context of Gaussian processes, as covariance functions.

In addition to the *Gaussian kernel* of Eq. (4.4), two further examples are the *Wiener* and *Ornstein–Uhlenbeck (exponential)* kernels over the reals (more in §5.4 and §5.5):

$$\begin{aligned} k_{\text{Wiener}}(a, b) &= \min(a, b), \\ k_{\text{Ornstein–Uhlenbeck}}(a, b) &= \exp(-|a - b|). \end{aligned} \quad (4.5)$$

There are other limits of certain feature sets that give rise to other kernels beyond the Gaussian form of Eq. (4.4) (Exercise 4.3). These constructions are not unique – there may be other choices of ostensibly very different feature sets that, in some limit, converge to the same covariance function. Nevertheless, this explicit construction provides two insights:

¹⁰ Proof sketch, omitting technicalities: Using Eq. (3.4), we get

$$\begin{aligned} k_{ab} &= \frac{\sqrt{2}\theta^2(c_{\max} - c_{\min})}{\sqrt{\pi}\lambda F} \\ &\quad \cdot \sum_{i=1}^F e^{-\frac{(a-c_i)^2}{\lambda^2}} e^{-\frac{(b-c_i)^2}{\lambda^2}} \\ &= \frac{\sqrt{2}\theta^2(c_{\max} - c_{\min})}{\sqrt{\pi}\lambda F} e^{-\frac{(a-b)^2}{2\lambda^2}} \\ &\quad \cdot \sum_{i=1}^F e^{-\frac{(c_i - 1/2(a+b))^2}{\lambda^2/2}}. \end{aligned}$$

In the limit of large F , the number of features in a region of width δc converges to $F \cdot \delta c / (c_{\max} - c_{\min})$, and the sum becomes the Gaussian integral

$$\begin{aligned} k_{ab} &= \frac{\sqrt{2}\theta^2}{\sqrt{\pi}\lambda} e^{-\frac{(a-b)^2}{2\lambda^2}} \\ &\quad \cdot \int_{c_{\min}}^{c_{\max}} e^{-\frac{(c - 1/2(a+b))^2}{\lambda^2/2}} dc. \end{aligned}$$

For $c_{\max}, c_{\min} \rightarrow \pm\infty$, that integral converges to $\sqrt{\pi}\lambda / \sqrt{2}$.

¹¹ It follows from the definition of positive definite matrices (§15.2) that all kernels obey $k(a, a) \geq 0$ and $k(a, b) = k(b, a)$, $\forall a, b \in \mathbb{X}$.

Exercise 4.3 (moderate). Convince yourself that similar limits for the “switches” and “steps” feature functions in Figure 4.2 give rise to the “linear splines” and Wiener kernels listed in Figure 4.3. More examples of this form, including infinite limits of polynomial and Fourier features, can be found in a technical report by Minka (2000).

- ⊖ While the covariance function (4.4) is a “sum” (an integral) over an *infinite* set of features, we had to drop the prior variance of each feature proportionally to M^{-1} , the number of features.¹² In this sense, we have spread a finite probability mass over an infinite space of features. The model may have infinitely many degrees of freedom, but not necessarily infinite modelling flexibility.
- ⊖ Constructing the kernel requires careful choices, like the scalar form of Σ , and feature functions with a well-defined and analytic integral limit. So the elegance of the nonparametric formulation comes at a price: while we can take *any* set of feature functions to build a parametric model, we cannot just take any bivariate function and hope that it might be a kernel.

¹² Over a bounded domain $[c_{\min}, c_{\max}]$, the determinant of this Σ , in the limit of $M \rightarrow \infty$, converges to the finite value

$$\begin{aligned}\det \Sigma &= M \cdot \frac{\sqrt{2}\theta^2}{\sqrt{\pi\lambda M}} (c_{\max} - c_{\min}) \\ &= \frac{\sqrt{2}\theta^2}{\sqrt{\pi\lambda}} (c_{\max} - c_{\min}).\end{aligned}$$

For an intuition of how large the space of positive definite kernels is, we note that kernels form a semi-ring.

1. If k is a kernel, then αk for any $\alpha \in \mathbb{R}_+$ is a kernel.

Proof. $v^\top K v > 0 \Rightarrow \alpha v^\top K v > 0$. □

2. If k, h are both kernels, then $k + h$ is a kernel.

Proof. $v^\top (K + H) v = v^\top K v + v^\top H v > 0$. □

3. If k, h are both kernels, then their Hadamard product $k \odot h$, i.e. the function

$$(k \odot h)(a, b) = k(a, b) \cdot h(a, b)$$

is a kernel. This result is known as *Schur's product theorem* and is significantly less straightforward than the former two. A proof can be found in Bapat (1997).

4. If $\phi : \mathbb{Y} \rightarrow \mathbb{X}$ is *any* function over a space $\mathbb{Y}$, then $k(\phi(y), \phi(y'))$ is a kernel over $\mathbb{Y}$. In particular, $k(x/s, x'/s)$ is a kernel for any linear *scale* $s \in \mathbb{R}_+$.

These observations show that, although not quite as unconstrained as the choice of features for parametric regression, the space of kernels is still quite large. Among the key results of subsequent chapters is the insight that certain classic numerical algorithms can be interpreted as performing regression using

a particular kernel. These kernels will often be associated with rather general, unstructured priors, and it will be a natural question whether more specific priors can be designed to better address individual numerical problems. That is, to what degree additional structural knowledge can help reduce the complexity of a computation.

▷ 4.2.2 Gaussian Process Regression

If kernels provide an infinite-dimensional extension of positive definite matrices, what does it mean to perform linear regression (in the sense of Algorithms 4.1 and 4.2) with a kernel? The probability measure implied by this use of a positive definite kernel in a covariance model is called a Gaussian process.

Definition 4.4 (Gaussian process). *Consider a function $\mu : \mathbb{X} \rightarrow \mathbb{R}$ and a positive definite kernel $k : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$. The Gaussian process (GP) $p(f) = \mathcal{GP}(f; \mu, k)$ is the probability measure identified by the property that, for any finite subset $X := [x_1, \dots, x_N] \subset \mathbb{X}$, the probability assigned to function values $f_X = [f(x_1), \dots, f(x_N)]$ is given by the multivariate Gaussian probability density $p(f_X) = \mathcal{N}(f_X; \mu_X, k_{XX})$.¹³*

The extension from parametric to nonparametric regression follows directly from Eq. (4.3). Given a Gaussian process prior $p(f) = \mathcal{GP}(f; \mu, k)$ over an unknown function $f : \mathbb{X} \rightarrow \mathbb{R}$, and the likelihood function from Eq. (4.2) (i.e. $p(\mathbf{y} | f) = \mathcal{N}(\mathbf{y}; f_X, \Lambda)$), the posterior over f is a Gaussian process $p(f | \mathbf{y}) = \mathcal{GP}(f; m, \mathbb{V})$ with mean function and SPD function, $m : \mathbb{X} \rightarrow \mathbb{R}$ and $\mathbb{V} : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$, respectively, of the form

$$m_x = \mu_x + k_{xX}(k_{XX} + \Lambda)^{-1}(\mathbf{y} - \mu_X), \quad (4.6)$$

$$\mathbb{V}(x, x') = k_{xx'} - k_{xX}(k_{XX} + \Lambda)^{-1}k_{Xx'}. \quad (4.7)$$

Figure 4.3 shows nine different Gaussian process priors over $\mathbb{X} = \mathbb{R}$, alongside the posterior measures arising from the non-linear noisy data set in Figure 4.1. The plots show the diversity of the Gaussian process hypothesis space. GP models can be stationary or spatially varying, and may produce smooth or rough sample paths. The combination of basic kernels and features – using the combination rules outlined in §4.2.1 – provides an extensive toolbox for the construction of prior models with specific properties.

Algorithms 4.3 and 4.4 provide a compact basic implementation for Gaussian process regression. Like Algorithms 4.1 and 4.2,

¹³ It is not straightforward to show that Definition 4.4 is well-formed, and what restrictions have to be put on the form of the kernel to guarantee the existence of such a family of infinitely many Gaussian random variables. For our purposes, it suffices to require the kernel to be continuous on both variables to guarantee the existence of the Gaussian process. Some hints can be found on p. 1 of Wahba's book (1990) and in Adler's (1990) in-depth treatment.

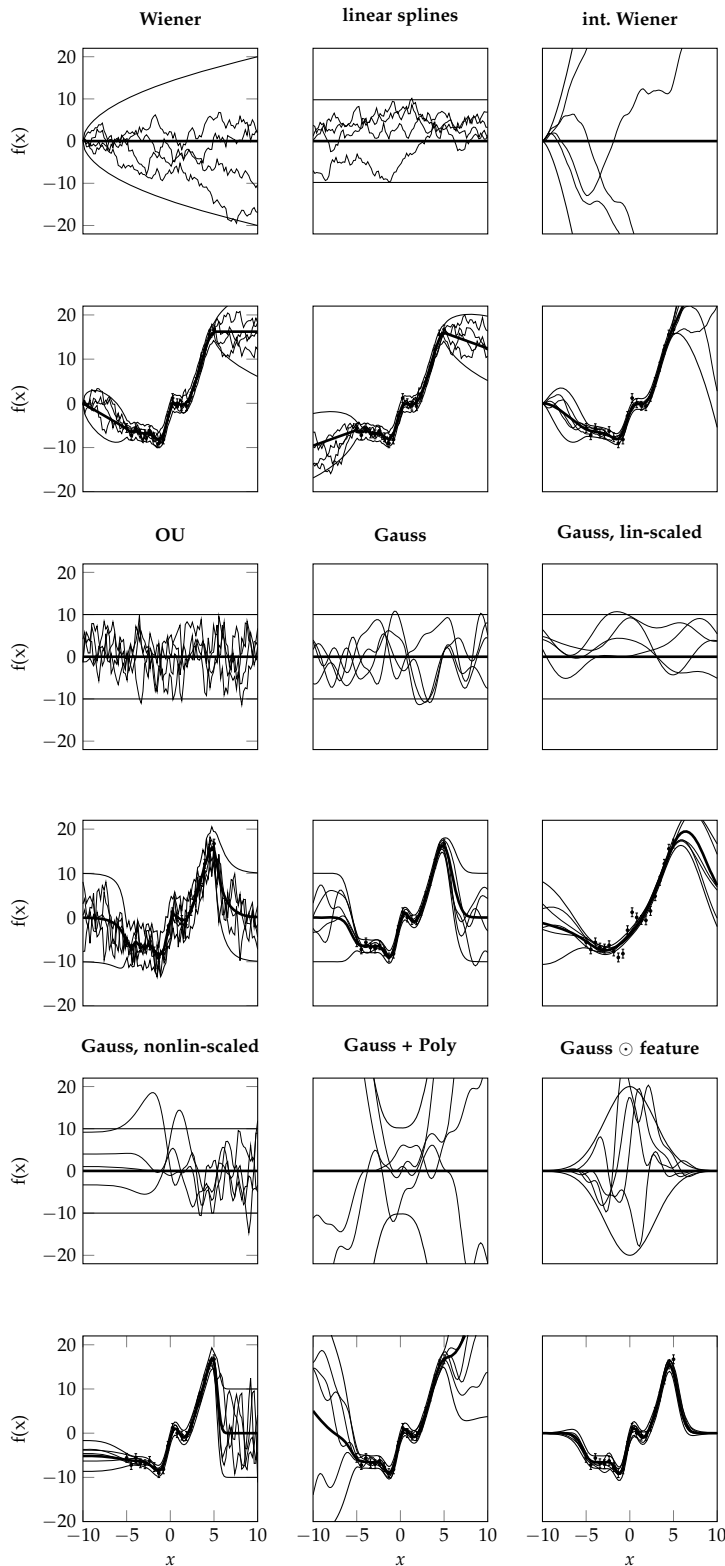


Figure 4.3: Analogous plot to Figure 4.2, for Gaussian process regression with various kernels. From top left to bottom right, the prior processes are identified by the following kernels: the Wiener kernel (producing Brownian motion)

$$k(a, b) = \min(a, b) + c, \quad c = 10;$$

the linear spline kernel

$$k(a, b) = 1 + c - \frac{c}{10} |a - b|, \quad c = 1;$$

the integrated Wiener kernel (using $\tilde{a} := a - 10, \tilde{b} := b - 10$)

$$k(a, b) = 1/3 \min(\tilde{a}, \tilde{b})^3 + 1/2 |\tilde{a} - \tilde{b}| \min(\tilde{a}, \tilde{b})^2;$$

the exponential kernel (Ornstein–Uhlenbeck process)

$$k(a, b) = e^{-|a-b|};$$

the Gaussian (square exponential, radial basis function) kernel

$$k_{SE} = e^{-(a-b)^2};$$

its linearly and nonlinearly scaling

$$k(a, b) = k_{SE}(\phi(a), \phi(b))$$

for the choices

$$\phi(x) = \frac{x}{3} \text{ and } \phi(x) = \left(\frac{(x+10)}{5} \right)^3;$$

the additive combination

$$k(a, b) = k_{SE}(a, b) + \sum_{i=0}^2 (ab)^i;$$

and the point-wise product

$$k(a, b) = k_{SE}(a, b) \cdot e^{-(a^2+b^2)/4}.$$

As in Figure 4.2, these examples demonstrate some of the variability among Gaussian process priors. Some processes (e.g. Wiener, spline, OU) produce sample paths that are almost surely nowhere differentiable, while others are smooth (e.g. the Gaussian kernel). Some are mean-reverting and stationary, others diverge.

```

1 procedure GP_INFER( $y, X, \Lambda, k, m$ )
2    $k_{XX} = k(X, X)$  // kernel at data
3    $m_X = m(X)$  // mean function at data
4    $G = k_{XX} + \Lambda$  // data covariance
5    $R = \text{CHOL}(G)$  // Cholesky decomp. of data covariance
6   // (i.e.  $G = R^T R$ ). This is the most costly step, at  $\mathcal{O}(N^3)$ .
7    $\alpha = R^{-1}(R^{-1})^T(y - m_X)$  // predictive weights
8 end procedure

```

they are separated into an *inference* and a *prediction step*. The computational load for inference on the GP posterior arising from N observations is dominated by the $\mathcal{O}(N^3)$ steps for a matrix square root of the so-called *Gram matrix* G – the algorithm uses the Cholesky decomposition, but other decompositions, such as eigendecompositions, could also be used in principle. The main point of such decompositions here is that they bring the matrix into a form from which linear systems can be solved in $\mathcal{O}(N^2)$ time. The cost of computing the mean function (the *mean prediction*) at a new location x_* is linear in N , computing the marginal variance at such a point is quadratic in N .

► 4.3 Relationship to Least-Squares and Statistical Learning

In the case of Gaussian models, the probabilistic formalism is closely connected to other frameworks for inference, learning and approximation. This connection is helpful to connect Probabilistic Numerics with classic numerical point estimates.

A theorem due to Moore and Aronszajn¹⁴ shows that each positive definite kernel is associated with a space of functions known as the *reproducing kernel Hilbert space* (*reproducing kernel Hilbert space* (RKHS)), defined through two abstract properties:¹⁵

```

1 procedure GP_PREDICT( $x, \alpha, R, m$ ) // predict at  $x \in \mathbb{X}^D$ 
2    $k_{xx} = k(x, x)$  // prior predictive covariance
3    $k_{xX} = k(x, X)$  // cross covariance
4    $m_x = m(x) + k_{xX}\alpha$  // posterior predictive mean
5    $L = k_{xX}/R$  // projection from data cov. to predictive space
6    $V_x = k_{xx} - LL^T$  // predictive covariance
7    $s_x = \text{CHOL}(V_x)^T \text{RANDNORMAL}(D) + m_x$  // samples
8 end procedure

```

Algorithm 4.3: Basic implementation of nonparametric Gaussian process *inference*. In contrast to Algorithm 4.1, inference must be performed in the function space, and has cost $\mathcal{O}(N^3)$. Line 7 re-uses the upper triangular matrix R for efficiency, because linear problems with such matrices can be solved at cost $\mathcal{O}(N^2)$ by back-substitution.

Exercise 4.5 (easy). Consider two functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and $g : \mathbb{R}^d \rightarrow \mathbb{R}$, both drawn from independent Gaussian processes as $f \sim \text{GP}(\mu_f, k_f)$ and $g \sim \text{GP}(\mu_g, k_g)$. From the definition of GPs it follows that the basic Gaussian property of Eq. (3.4) carries over: If A is a (sufficiently regular) linear operator and $p(x) = \mathcal{GP}(x; m, k)$, then $p(Ax) = \mathcal{N}(Ax; Am, AkA^T)$. Use this property to answer the following questions.

1. Consider the real number $\alpha \in \mathbb{R}$. The function $\tilde{f} = \alpha \cdot f$ is also distributed according to a Gaussian process. What are its mean and covariance functions?
2. The sum $f + g$ is also distributed according to a Gaussian process. What is the mean function and the kernel of $f + g$?
3. The sum $f - g$ is also distributed according to a Gaussian process. What is the mean function and the kernel of $f - g$?

¹⁴ This theorem was first published by Aronszajn (1950), but he attributed it to E. H. Moore. A proof can also be found in Wahba's book (1990, Thm. 1.1.1).

¹⁵ Schölkopf and Smola (2002), Def. 2.9.

Algorithm 4.4: Basic implementation of nonparametric Gaussian process *prediction*. Predicting at D input locations has cost $\mathcal{O}(D^2)$ for a full covariance, and $\mathcal{O}(D)$ for a mean prediction and marginal (element-wise) variance. In the latter case, line 7 is left out, and the diagonal of V_x in line 6 can be computed cheaply using $[LL^T]_{ii} = \sum_j [L_{ij}]^2$. Compare with Algorithm 4.2.

Definition 4.6. A Hilbert space $\mathcal{H}$ of functions $f : \mathbb{X} \rightarrow \mathbb{R}$ is called an RKHS if there exists a function $k : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ such that

1. for every $x \in \mathbb{X}$, the function $f(\tilde{x}) = k(x, \tilde{x})$ is in $\mathcal{H}$, and
2. k has the reproducing property that the values of every $f \in \mathcal{H}$ at any $x \in \mathbb{X}$ can be written as $f(x) = \langle f(\cdot), k(\cdot, x) \rangle$ (where $\langle \cdot, \cdot \rangle$ is the inner product of $\mathcal{H}$).

Using the RKHS notion the GP posterior mean estimate m_x of Eq. (4.6) can equivalently be derived as a nonparametric least-squares estimate, also known as the *kernel ridge regression* estimate: m_x equals the minimiser of a *regularised empirical risk* functional over the RKHS associated with k :¹⁶

$$\begin{aligned} m_x &= k_{xX}(k_{XX} + \sigma^2 I)^{-1} \mathbf{y} \\ &= \arg \min_{f \in \mathcal{H}_k} \sigma^2 \|f\|_{\mathcal{H}}^2 + \frac{1}{N} \sum_{i=1}^N (y_i - f(x_i))^2, \end{aligned} \quad (4.8)$$

where $\|f\|_{\mathcal{H}}$ is the RKHS norm of f .¹⁷ The corresponding statement for the posterior standard deviation

$$\sigma(x) := \sqrt{\mathbb{V}(x, x)} = \sqrt{k_{xx} - k_{xX}(k_{XX} + \Lambda)^{-1} k_{Xx}} \quad (4.9)$$

is given by the following theorem about the *worst-case approximation error* of ridge regression. For simplicity, we focus on the special case of function values observed without noise, which is most relevant for classic numerical problems.¹⁸

Theorem 4.8. Let k be a positive definite kernel over $\mathbb{X}$, and $\mathcal{H}$ its associated RKHS. Consider a function $f \in \mathcal{H}$ with $\|f\|^2 \leq 1$, and a finite subset $X \in \mathbb{X}^N$. Denote $\mathbf{y} = f_X \in \mathbb{R}^N$. Then, for any $x \in \mathbb{X}$, the approximation error between $f_x := f(x) \in \mathbb{R}$ and $m_x \in \mathbb{R}$ as defined by Eq. (4.6) is tightly bounded above by the associated GP posterior marginal variance for $\Lambda \rightarrow 0$:

$$\sup_{f \in \mathcal{H}, \|f\| \leq 1} (m_x - f_x)^2 = k_{xx} - k_{xX} K_{XX}^{-1} k_{Xx}.$$

This statement shows that in the case of error-free evaluations of f , the posterior variance of Gaussian process regression – the Probabilists' *expected error* – equals the *worst-case approximation error* if the true function is an element of the RKHS of unit-bounded norm.¹⁹

Thus, if we are faced with the problem of inferring (estimating, approximating) a function $f : \mathbb{X} \rightarrow \mathbb{R}$ from observations $(x_i, y_i)_{i=1, \dots, N}$, we now have two quite different viewpoints on the very same point- and error-estimates:

¹⁶ A proof of this theorem can be found in Kanagawa et al. (2018), a paper that also contains many other results on the connection between the GP and kernel formalisms. The proof requires an appeal to additional results, which are left out here for brevity.

¹⁷ Note that, in this text, the meaning of the symbol k is overloaded. On the one hand, it signifies the covariance function of a *Gaussian process* (GP) and, on the other hand, the kernel of an RKHS. This double use of k is common practice due to the similarity of both cases.

¹⁸ For the general case of $\Lambda \rightarrow 0$, see Kanagawa et al. (2018), §3.4.

Exercise 4.7 (moderate, solution on p. 359). Prove Theorem 4.8. Hint: Use the reproducing property from Definition 4.6, and the Cauchy–Schwarz inequality

$$|\langle f, g \rangle|^2 \leq \langle f, f \rangle \cdot \langle g, g \rangle,$$

for $f, g \in \mathcal{H}$.

¹⁹ For clarity: The posterior standard-deviation σ_x of Eq. (4.9), in general, is not itself an element of the RKHS.

1. Assign a Gaussian process prior $p(f) = \mathcal{GP}(f; 0, k)$ over a hypothesis class of functions and use the asymptotic Gaussian likelihood $p(y \mid f) = \prod_{i=1}^N \delta(y_i - f(x_i))$. The resulting posterior is a Gaussian process with mean m_x and marginal variance σ_x .
2. Assume that the true function f is an element of the RKHS $\mathcal{H}_k$ associated with k , and decide to construct the regularised least-squares estimate for f within $\mathcal{H}_k$ (Eq. (4.8)), which is given by m_x . The distance between this estimate and the true $f(x)$ (if the assumption $f \in \mathcal{H}_k$ is correct!) is bounded above, up to an unknown constant (the RKHS-norm of f) by σ_x .

In many sections of this text, this equivalence between probabilistic and statistical estimation will be used to construct probabilistic re-interpretations of classic numerical methods. The juxtaposition above highlights at least one reason why it is helpful to have access to both formulations, and thus why it makes sense to study Probabilistic Numerics in addition to classic methods: in contrast to an empirical risk functional, a prior is a generative object. We can study draws from the prior to analyse and criticise the assumptions encoded in the prior. If a particular GP prior does not encode certain properties we tangibly know about our concrete problem, we may change the prior to better reflect our knowledge. It is much harder to do so based on an empirical risk. We also note that while the Gaussian inference formulation adds this additional conceptual strength, it does not take something away from the statistical one: Both in terms of intrinsic assumptions and computational demands, both are equivalent to each other.

There are, however, also subtle differences between the two formulations. In particular, draws from a Gaussian process are in general not elements of the RKHS.²⁰ Instead they come for a different space that, in general, is slightly “larger” and “rougher” surrounding the RKHS. This aspect will have tangible consequences for the error analysis of probabilistic numerical methods. In §II and later, we will see that probabilistic error estimates of *classic* methods are usually overly cautious on the problems that these methods are usually applied to, an indication that these methods are too generic, and can be improved upon with additional prior information.

²⁰ See p. 131 in Rasmussen and Williams (2006). Detailed analyses are offered by Driscoll (1973) and, more recently, Steinwart (2019). There is also a detailed discussion in §4 of Kanagawa et al. (2018).

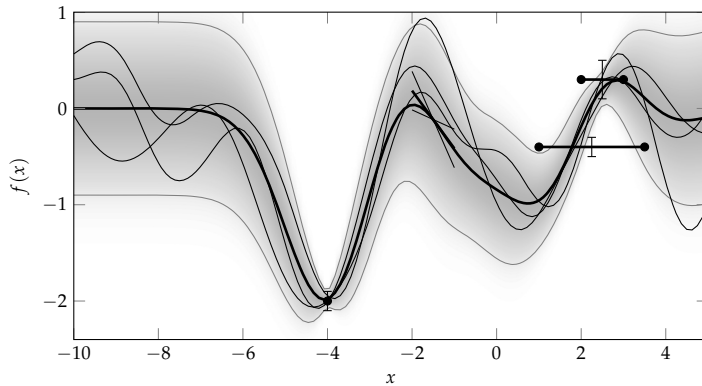


Figure 4.4: Gaussian process inference on a function f from evaluations of the function ($f(x = -4)$), its derivative ($f'(x = -1.5)$) and integrals ($\int_a^b f(x) dx$ for $[a, b] = [1, 3.5]$ and $[2, 3]$, plotted are the average function values, the value of the integral divided by $(b - a)$). Each observation is corrupted by Gaussian noise of varying degree (plotted error bars). These observations have a structured effect on the uncertainty assigned to function values. For example, the derivative observation constrains the derivatives of f , but contains only weak information about the absolute deviation of f from the mean function.

► 4.4 Inference from and on Derivatives and Integrals

The closure of Gaussian distributions under linear projections (Eq. (3.4)) allows for an elegant extension of regression on functions from observed function values, which is of great importance for numerical uses of the Gaussian inference framework. Consider a real-valued function over a Euclidean vector space, $f : \mathbb{R}^N \rightarrow \mathbb{R}$, and assign the Gaussian process prior $p(f) = \mathcal{GP}(f; m, k)$. Assume that the mean function $m(x)$ and the kernel $k(x, x')$ are at least q -times continuously differentiable in all their arguments, and integrable against the measure $\nu : \mathbb{R}^N \rightarrow \mathbb{R}$. Then it follows²¹ from Eq. (3.4) that all partial derivatives and integrals against ν are jointly Gaussian (process) distributed, with mean functions

$$\mathbb{E} \left(\frac{\partial^\ell f(x)}{\partial x_i^\ell} \right) = \frac{\partial^\ell m(x)}{\partial x_i^\ell} \quad \text{for } 0 \leq \ell \leq q,$$

$$\mathbb{E} \left(\int_a^b f(x) d\nu(x) \right) = \int_a^b m(x) d\nu(x),$$

and covariance functions of the general form

$$\text{cov} \left(\frac{\partial^\ell f(x)}{\partial x_i^\ell}, \frac{\partial^m f(x')}{\partial x_j^m} \right) = \frac{\partial^\ell \partial^m k(x, x')}{\partial x_i^\ell \partial x_j^m},$$

$$\text{cov} \left(\int_a^b f(x) d\nu(x), \frac{\partial^m f(x')}{\partial x_j^m} \right) = \int \frac{\partial^m k(x, x')}{\partial x_j^m} d\nu(x),$$

and analogously for mixed partial derivatives and higher-order integrals. Of course, these results can be used to perform inference on integrals from derivative observations and function values, and the other way round. This tool will be a staple of probabilistic numerical methods derived in later chapters.

²¹ There are subtleties involved due to the nonparametric nature of the Gaussian process measure, so the appeal to Eq. (3.4) is simplistic, made with practicality in mind. A more precise statement is that, if L is a linear operator acting on f , then Lf is a Gaussian process with mean function Lm and covariance function LkL^* , if Lm and $Lk(\cdot, x')$ are bounded. The specific result for derivatives can be found on p. 27 in Adler (1981). For a technical treatment, see Akhiezer and Glazman (2013). Thanks to Simo Särkkä for pointing out these references. A more recent discussion is offered by Owhadi and Scovel (2015).

The kernels used in this context in later chapters include, most prominently, the Wiener process and its integrals (see §5.4). Also in wide use, mostly for algebraic convenience, is the Gaussian (aka square-exponential, radial basis function) kernel

$$k_{\text{SE}}(a, b) := \exp\left(-\frac{(a-b)^2}{2\lambda^2}\right).$$

The Gaussian kernel amounts to an extreme assumption of smoothness on f , but its extremely convenient algebraic properties nevertheless make it a versatile tool for rather regular problems, if extreme numerical precision is not the primary goal. For easy reference, we include explicit forms for the most frequent combinations of partial derivatives and integrals here, using the shorthand $\Phi(z) := \text{erf}\left(\frac{z}{\sqrt{2}\lambda}\right)$. If $p(f) = \mathcal{GP}(f; 0, k_{\text{SE}})$, then

$$\begin{aligned} \text{cov}\left(\int_a^b f(x)dx, \int_c^d f(\tilde{x})d\tilde{x}\right) &= \int_a^b \int_c^d k_{\text{SE}}(x, \tilde{x}) dx d\tilde{x} \\ &= \sqrt{\frac{\pi\lambda^2}{2}} \left[-(d-b)\Phi(d-b) + (d-a)\Phi(d-a) \right. \\ &\quad \left. + (c-b)\Phi(c-b) - (c-a)\Phi(c-a) \right] \\ &\quad + \lambda^2 \left[-k_{\text{SE}}(d, b) + k_{\text{SE}}(d, a) + k_{\text{SE}}(c, b) - k_{\text{SE}}(c, a) \right], \\ \text{cov}\left(\int_a^b f(\tilde{x})d\tilde{x}, f(x)\right) &= \int_a^b k_{\text{SE}}(x, \tilde{x}) d\tilde{x} = \text{cov}\left(f(x), \int_a^b f(\tilde{x})d\tilde{x}\right) \\ &= \sqrt{\frac{\pi\lambda^2}{2}} \left[\Phi(b-x) - \Phi(a-x) \right], \\ \text{cov}\left(\int_a^b f(\tilde{x})d\tilde{x}, \frac{\partial f(x)}{\partial x}\right) &= \frac{\partial}{\partial x} \int_a^b k_{\text{SE}}(x, \tilde{x}) d\tilde{x} = \text{cov}\left(\frac{\partial f(x)}{\partial x}, \int_a^b f(\tilde{x})d\tilde{x}\right) \\ &= k_{\text{SE}}(x, b) - k_{\text{SE}}(x, a), \end{aligned}$$

and

$$\begin{aligned} \text{cov}\left(f(x), \frac{\partial f(\tilde{x})}{\partial \tilde{x}}\right) &= \frac{\partial k_{\text{SE}}(x, \tilde{x})}{\partial \tilde{x}} = -\text{cov}\left(\frac{\partial f(x)}{\partial x}, f(\tilde{x})\right) \\ &= \frac{x-\tilde{x}}{\lambda^2} k_{\text{SE}}(x, \tilde{x}), \\ \text{cov}\left(\frac{\partial f(x)}{\partial x}, \frac{\partial f(\tilde{x})}{\partial \tilde{x}}\right) &= \frac{\partial^2 k_{\text{SE}}(x, \tilde{x})}{\partial x \partial \tilde{x}} \\ &= \left(\frac{1}{\lambda^2} - \frac{(x-\tilde{x})^2}{\lambda^4} \right) k_{\text{SE}}(x, \tilde{x}), \\ \text{cov}\left(\frac{\partial^2 f(x)}{\partial x^2}, \int_a^b f(\tilde{x})d\tilde{x}\right) &= \frac{1}{\lambda^2} \left[(x-b)k_{\text{SE}}(x, b) - (x-a)k_{\text{SE}}(x, a) \right] \\ \text{cov}\left(\frac{\partial^2 f(x)}{\partial x^2}, f(\tilde{x})\right) &= \left(\frac{(x-\tilde{x})^2}{\lambda^4} - \frac{1}{\lambda^2} \right) k_{\text{SE}}(x, \tilde{x}). \end{aligned}$$